AnalogRetriever: Learning Cross-Modal Representations for Analog Circuit Retrieval

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Research Questions

- How can we enable cross-modal retrieval across heterogeneous analog circuit representations (SPICE netlists, schematics, text descriptions)?
- How can we align graph-structured netlists with visual schematics and textual descriptions in a unified embedding space?
- How can retrieval-augmented generation improve LLM-based analog circuit design?

Core Idea

Analog circuit designers currently search for reference circuits manually across fragmented sources (papers, manuals, databases) using limited keyword matching within single modalities. The key insight is that the same circuit exists as three different representations—SPICE netlist (graph code), schematic (image), and functional description (text)—that should be semantically aligned. AnalogRetriever uses a port-aware relational GCN for netlists (capturing device terminal semantics), pretrained CLIP for images/text, and maps all three into a shared embedding space via contrastive learning with curriculum training.

Key Formula

Port-aware RGCN message passing

$$h_v^{(l+1)} = \sigma\left(\sum_{r \in R} \sum_{u \in N_r(v)} \frac{1}{|N_r(v)|} W_r^{(l)} h_u^{(l)} + h_v^{(l)}\right)$$

Where N_r(v) is neighbors of node v under relation type r, W_r^{(l)} is relation-specific weight matrix for edge type r at layer l, enabling the model to distinguish different device terminals (gate vs drain vs source) which is critical for analog circuit topology understanding

Curriculum hard-negative ratio schedule

$$\alpha_m = \min(\alpha_{max}, \alpha_0 + \frac{m-1}{M-1}(\alpha_{max} - \alpha_0))$$

Where m is current epoch in Phase 3, M is total Phase-3 epochs, α_0=0.05 is initial ratio, α_max=0.3 is maximum ratio. Progressively increases fraction of hard negatives from same functional cluster to sharpen discrimination between topologically distinct circuits with similar descriptions

Key Figure

synthesis. AnalogXpert [19] formulates topology synthesis as subcircuit-level SPICE generation, AnalogSeeker [20] builds a domain-specific foundation model, LaMAGIC [3] casts topology generation as language modelling, Artisan [4] automates op-amp design end-to-end, and AnalogGenie [5] explicitly explores the topology space. These methods focus on circuit *generation* but suffer from hallucination and invalid topologies. Our retrieval component grounds LLM-based tools with relevant circuit examples via RAG, reducing invalid outputs.

2.3 Cross-Modal Retrieval and Contrastive Learning

Contrastive vision-language pretraining [12, 13] has become the de facto approach for aligning heterogeneous modalities, but its direct application to circuit data is fundamentally limited: SPICE netlists are graph-structured with arbitrary node names, and circuits sharing similar descriptions can differ substantially at the device level. Graph neural networks for EDA [17, 21, 22] recover structural semantics from netlists but are typically trained in isolation, without alignment to natural language or schematics. To our knowledge, no prior work establishes a single representation space that bridges all three analog circuit modalities; AnalogRetriever closes this gap and supports all six cross-modal directions with a

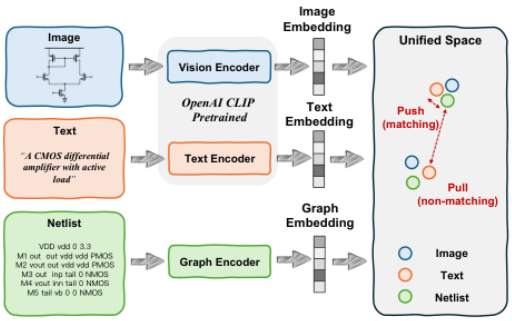

Figure 2. AnalogRetriever framework. Three modality-specific encoders map SPICE netlists (port-aware RGCN), schematic images (ViT), and text descriptions (Transformer) into a shared embedding space, trained with tri-modal contrastive learning, auxiliary circuit-type classification, and three-phase curriculum with hard-negative mining.

shared d=768 space. To preserve the pretrained cross-modal alignment while allowing domain adaptation to clean circuit line drawings, we freeze the bottom 16 of 24 ViT blocks and fine-tune only the top 8, avoiding catastrophic forgetting of CLIP's general visual-semantic prior.

Figure 2: Shows the complete AnalogRetriever architecture with three modality-specific encoders (CLIP ViT for images, CLIP Transformer for text, port-aware RGCN for netlists) mapping into shared embedding space via contrastive learning

Limitations

- Dataset limited to 19 canonical analog topologies, excluding mixed-signal, RF, and power-management circuits
- Uses 20 hand-defined port types for RGCN rather than learning relation vocabulary from data
- Requires efficient nearest-neighbor search methods for industrial-scale databases
- Query expansion via LLM rewriting needed for short task prompts to achieve good retrieval performance